Parsa's Thesis

First Meeting — Research Ideas & Overview

Parsa

October 19, 2026

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Research Roadmap

- **Project Seminar**
- **RQs**
  - RQ1 — Prompt Injection & Horizon
  - RQ2 — Vector DB
  - RQ3 — Rogue AI

Architecture & Evaluation

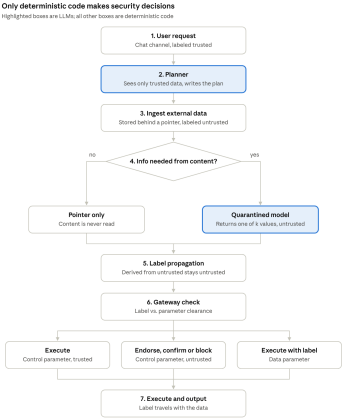
- **BlackHole**
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RQ1 — Prompt Injection

- Conformal Predictions
- Certified Robustness
- Pearson inequality

RQ1 — Horizon

Only deterministic code makes security decisions; LLMs remain isolated behind strict labels and pointers.



RQ2 — Vector DB

Differential Privacy
To prevent Data Leakage

Conformal Methods and bounds for poisoning

DP-SGD concept

GraphRAG relations poisoning

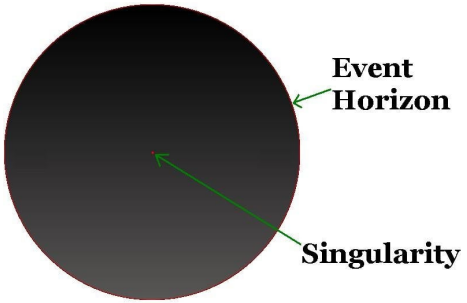
RQ 3 — Rogue AI

Monitoring

- **Probability bounds for incoming malicious tokens:** This shall be over time, via Importance Sampling + extreme Power Law + Estimate Deployments scale
- **SPRT + E-values**
- **Ville's Inequality** (Vitllie Inequality)
- **Rare Event estimations**

Project Blackwall upgrade

BlackHole


Event
Horizon
Singularity
Schwarzschild radius
$R=\frac{2GM}{c^2}$

Game-Theoretic Formulation

Sion Minimax Game

Evaluations

DeepTeam

AgentDojo (AgentDoji)

Hoeffding intervals

Summary

1. **Scope**: indirect (and direct) prompt injection in RAG/agent pipelines
2. **Spine**: IFC + noninterference with a proof, enforced by (a) **grammar-constrained decoding** that gives unauthorized tool calls zero probability, and (b) a **reference monitor that kills on violation** (Schneider: safety properties are enforceable)
3. **Probabilistic barriers**: perplexity filter, classifier, reasoning/task-alignment layer, **decoding-time suppression** (SafeDecoding/DExperts-style), optionally circuit breakers
4. **Internal signals → kill**: suppressed probability mass, entropy spike, attention shift (Attention Tracker)
5. **Game**: finite mixed-strategy minimax over barrier configs × attack classes; kills and false positives in the payoff
6. **Worst case**: game value capped by the IFC/monitor bound
7. **Evaluation**: DeepTeam + AgentDojo on open-weights models; attack success rate and utility, with Hoeffding interval
8. **Parking lot**: smarter thinking layer, edit automata

Techs to use

Nvidia

- Openshell
- Model Optimizer
- Garak
- Guardrails

Microsoft

- Spotlighting
- GraphRAG

Google

- RRSI & Dream-RSI
- VeriHarness
- CaMeL

Tooling & Open-Source

- Adversarial Robustness Toolbox (ART) v1.20
- Unstract & LLMWhisper
- Chrootkit (Chtrootkit)
- vLLM Guided Outline Decoding

Contact

Thank you!

Questions & Discussion